Maxwell's equation in free space

$$\rho = 0, \quad J = 0$$

Maxwell's equation of Harmonically varying field

$$D = D_0 e^{i\omega t} \quad \& \quad B = B_0 e^{i\omega t}$$

$$\nabla \cdot D = \rho$$

$$\nabla \cdot B = 0$$

$$\nabla \times E + i\omega B = 0$$

$$\nabla \times H - i\omega B = J$$

Poynting Theorem:- It states that "The net power flowing out of a given volume V is equal to the time rate of decrease in energy stored with in V minus the ohmic losses."

$$\underbrace{\oint \text{div} (\vec{E} \times \vec{H})\, dV}_{\text{Total power leaving the volume}} = \underbrace{-\frac{\partial}{\partial t} \int_V \left[\frac{1}{2}\epsilon E^2 + \frac{1}{2}\mu H^2\right] dV}_{\text{Rate of decrease in energy stored in electric and mag. field}} \underbrace{- \int \sigma E^2 dV}_{\text{ohmic power dissipated}}$$

Total power leaving the volume = Rate of decrease in energy stored in electric and mag. field − ohmic power dissipated

→ Electrostatic Pot. energy

$$U_e = \frac{1}{2} \int \vec{E} \cdot \vec{D}\, dV$$

magnetostatic energy

$$U_m = \frac{1}{2} \int \vec{H} \cdot \vec{B}\, dV$$

As we know

$$\vec{\nabla} \cdot \vec{D} = \rho \quad \text{--- (1)}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{--- (2)}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (3)}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{--- (4)}$$

Take scalar product of (3) with H and (4) with $\vec{E}$

$$H \cdot \text{Curl}\, E = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t}$$

$$E \cdot \text{Curl}\, H = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

→ Vector Identity: $\text{div}\,(\vec{E} \times \vec{H}) = H \cdot \text{curl}\, E - E\, \text{curl}\, H$

$$\underbrace{H \cdot \text{curl}\, E - E \cdot \text{Curl}\, H}_{\text{div}\,(\vec{E} \times \vec{H})} = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - E \cdot \frac{\partial \vec{D}}{\partial t} - \vec{E} \cdot \vec{J}$$

$$\text{div}\,(\vec{E} \times \vec{H}) = -\left(\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}\right) - \vec{E} \cdot \vec{J}$$

$$E \cdot \frac{\partial D}{\partial t} = E \cdot \frac{\partial}{\partial t}(\epsilon E) = \frac{1}{2} \epsilon \frac{\partial}{\partial t}(E)^2 = \frac{\partial}{\partial t}\left(\frac{1}{2} \vec{E} \cdot \vec{D}\right)$$

$$H \cdot \frac{\partial \vec{B}}{\partial t} = H \cdot \frac{\partial}{\partial t}(\mu H) = \frac{1}{2} \mu \frac{\partial}{\partial t}(H)^2 = \frac{\partial}{\partial t}\left(\frac{1}{2} H \cdot B\right)$$

$$\rightarrow \int \text{div}\,(\vec{E} \times \vec{H})\, dv = \int_V -\frac{\partial}{\partial t}\left[\frac{1}{2}(\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B})\right] dv - \int \vec{J} \cdot \vec{E}\, dv$$

$$= -\frac{\partial}{\partial t} \int \frac{1}{2}(\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B})\, dv - \int \sigma E^2 dv$$

$$\Rightarrow \int \text{div}\,(\vec{E} \times \vec{H})\, dv = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2\right) dv - \int_V \sigma E^2 dv$$

$\vec{E} \times \vec{H} = \vec{S}$ = Poynting vector or power flux or amount of energy crossing unit area placed perpendicular to the vector per unit time

It points along the dirn of flow of radiation.

Plane Electromagnetic waves in free space :-

As we know, Maxwell's equations are :

$$\vec{\nabla}\cdot\vec{D} = \rho$$
$$\vec{\nabla}\cdot\vec{B} = 0$$
$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
$$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

for free space

$\Rightarrow$ $\rho = 0$, $J = 0$, $\mu = \mu_0$ and $\epsilon = \epsilon_0$

$$\vec{\nabla}\cdot\vec{E} = 0 \quad \text{— (1)}$$
$$\vec{\nabla}\cdot\vec{H} = 0 \quad \text{— (2)}$$
$$\vec{\nabla}\times\vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{— (3)}$$
$$\vec{\nabla}\times\vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{— (4)}$$

taking curl of (3) & (4) equation.

→ from (3) $\Rightarrow$

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E} = -\mu_0 \frac{\partial \vec{\nabla}\times\vec{H}}{\partial t} = -\mu_0\epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\underbrace{\text{grad.}(\text{div } \vec{E})}_{0} - \nabla^2 E = -\mu_0\epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

$$\boxed{+\nabla^2 E = \mu_0\epsilon_0 \frac{\partial^2 E}{\partial t^2}}$$

or

$$\nabla^2 E - \mu_0\epsilon_0 \frac{\partial^2 E}{\partial t^2} = 0$$

from (4) similarly, we have

$$\nabla^2 H = \mu_0\epsilon_0 \frac{\partial^2 H}{\partial t^2}$$

or

$$\nabla^2 H - \mu_0\epsilon_0 \frac{\partial^2 H}{\partial t^2} = 0$$

In general

$$\nabla^2 \phi - \mu_0\epsilon_0 \frac{\partial^2 \phi}{\partial t^2} = 0 \quad \text{— (5)}$$

It is similar to general wave equation

$$\nabla^2 \phi = \frac{1}{v^2} \frac{\partial^2 \phi}{\partial t^2} \quad \text{— (6)}$$

By Comparing equation (5) & (6)

$$v^2 = \frac{1}{\mu_0\epsilon_0} \quad \text{or} \quad v = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

$\mu_0 = 4\pi \times 10^{-7}$ wb/A-m

$\epsilon_0 = 8.85 \times 10^{-12}$ F/m (farad/m)

$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ m/F

$\Rightarrow$ $v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8$ m/s $= c$

$\Rightarrow$ equations
$$\nabla^2 E - \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2} = 0$$
$$\nabla^2 H - \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2} = 0$$

Solution of plane wave

$$E(r,t) = \underset{\sim}{E_0}\, e^{i(\vec{K}\cdot\vec{r} - \omega t)}$$

Similarly $H(r,t) = \underset{\sim}{H_0}\, e^{i(\vec{K}\cdot\vec{r} - \omega t)}$

where E_0 & H_0 are complex amplitudes, K is wave propagation vector.

$\rightarrow \frac{\nu}{1} = \frac{c}{\lambda}$

So $\vec{K} = k.\hat{n} = \frac{2\pi}{\lambda}\hat{n}$

$\Rightarrow \frac{1}{\lambda} = \frac{\nu}{c}$

$= \frac{2\pi\nu}{c}\hat{n} = \frac{\omega}{c}\hat{n}$ → unit vector

Now for condition i.e solution must satisfies
$$\vec{\nabla}\cdot\vec{E} = 0$$

$$= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot E_0 e^{i(\vec{K}\cdot\vec{r} - \omega t)}$$

$$= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot\left(\hat{i}E_{0x} + \hat{j}E_{0y} + \hat{k}E_{0z}\right) e^{i(K_x x + K_y y + K_z z) - i\omega t}$$

$$= \left[E_{0x}\, i K_x + E_{0y}\, i K_y + E_{0z}\, i K_z\right] e^{i\vec{K}\cdot\vec{r} - i\omega t}$$

$$= i\,(k_x E_{0x} + k_y E_{0y} + k_z E_{0z})\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$$

$$= i\,(\hat{i}k_x + \hat{j}k_y + \hat{k}k_z)\cdot(\hat{i}E_{0x} + \hat{j}E_{0y} + \hat{k}E_{0z})\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$$

$$= i\,\vec{k}\cdot\vec{E_0}\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$$

$$= i\,\vec{k}\cdot\vec{E}$$

Similarly $\vec{\nabla}\cdot\vec{H} = i\,\vec{k}\cdot\vec{H}$

$i\vec{k}\cdot\vec{E} = 0$ or $i\vec{k}\cdot\vec{H} = 0$

$k \perp E$ or $k \perp H \Rightarrow \vec{E}$ and $\vec{H}$ are $\perp^r$ to direction of propagation

Similarly

$\vec{\nabla}\times\vec{H} = i\vec{k}\times\vec{H}$ or we have

$\vec{\nabla}\times\vec{E} = i\vec{k}\times\vec{E}$ $\quad \vec{\nabla}\times\vec{E} = -\mu_0 \dfrac{\partial\vec{H}}{\partial t}$

$i\vec{k}\times\vec{E} = (-i\omega H)(\mu_0)$

$$\boxed{\vec{k}\times\vec{E} = \omega\mu_0\vec{H}} \quad (7)$$

or $i\,\vec{k}\times\vec{H} = \epsilon_0(-i\omega\vec{E})$

$$\Rightarrow \boxed{\vec{k}\times\vec{H} = -\epsilon_0\omega\vec{E}} \quad 8$$

H and E both $\perp^r$ to each other

$$\vec{H} = \frac{1}{\omega\mu_0}\,\vec{k}\times\vec{E}$$

$$\vec{H} = \frac{1}{\omega\mu_0}\frac{\omega}{c}\,\hat{n}\times\vec{E} = \frac{1}{\mu_0 c}(\hat{n}\times\vec{E})$$

The equation in terms of modulus

$$H = \frac{1}{\mu_0 c} E$$

$$Z_0 = \left|\frac{\vec{E}}{\vec{H}}\right| = \left|\frac{\vec{E_0}}{H_0}\right| = \mu_0 c = \frac{\mu_0}{\sqrt{\mu_0 \epsilon_0}} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.6 \text{ ohm}$$
$$\approx 377 \text{ ohms}$$

wave impedance of free space

$\because$ $E =$ Volt/meter

$H =$ Ampere/meter

$$E/H = \frac{\text{Volt}}{\text{Ampere}} = \text{ohm}$$

$\Rightarrow \vec{E}$ and $\vec{H}$ are in same phase

→ Poynting vector

$$\vec{S} = \vec{E} \times \vec{H} = \frac{\vec{E} \times \hat{n} \times \vec{E}}{\mu_0 c} = \frac{1}{\mu_0 c} E^2 \hat{n}$$

energy flow per unit area per unit time.

$$(E \cdot E)\hat{n} - \underbrace{(E \cdot \hat{n})}_{0} E$$

$\because \vec{E} \cdot \hat{n} = 0 \quad E \perp \hat{n}$

$$\Rightarrow \boxed{\vec{S} = \frac{E^2}{Z_0} \hat{n}}$$

average of $\vec{S} = \langle \vec{S} \rangle = \frac{1}{Z_0} \langle \vec{E}^2 \rangle \hat{n}$

$$= \frac{1}{Z_0} \left\langle \left(E_0 e^{i\vec{k}\cdot\vec{r} - i\omega t} \right)^2 \right\rangle_{real} \hat{n}$$

$$= \frac{1}{Z_0} E_0^2 \langle \underbrace{\cos^2(\vec{k}\cdot\vec{r} - \omega t)}_{\frac{1}{2}} \rangle \hat{n}$$

$$= \frac{1}{Z_0} \frac{E_0^2}{2} \hat{n}$$

$\because E_{rms} = \frac{E_0}{\sqrt{2}}$

$$\langle \vec{S} \rangle = \frac{1}{Z_0} E_{rms}^2 \hat{n}$$

$\Rightarrow$ flow of energy in a plane E.m. wave in free space is along the direction of wave.

Ratio of electrostatic and magnetic energy densities

or $$\frac{u_e}{u_m} = \frac{\frac{1}{2}\epsilon_0 E^2}{\frac{1}{2}\mu_0 H^2} = \frac{\epsilon_0 E^2}{\mu_0 H^2} = \frac{\epsilon_0}{\mu_0} \cdot \frac{\mu_0}{\epsilon_0} = 1$$

total electromagnetic ~~energy~~ energy density

$$u = u_e + u_m = 2 \times \frac{1}{2}\epsilon_0 E^2 = \epsilon_0 E^2$$

$$\langle u \rangle = \langle \epsilon_0 E^2 \rangle = \epsilon_0 \langle (E_0 e^{ik\cdot r - i\omega t})^2 \rangle_{real}$$

$$\epsilon_0 E_0^2 \cdot \frac{1}{2} = \epsilon_0 E_{rms}^2$$

$$\frac{\langle S \rangle}{\langle U \rangle} = \frac{1}{Z_0} \frac{E_{rms}^2 \hat{n}}{\epsilon_0 E_{rms}^2} = \frac{\hat{n}}{Z_0 \epsilon_0} = \frac{\hat{n}}{\sqrt{\frac{\mu_0}{\epsilon_0}}\,\epsilon_0}$$

$$= \frac{\hat{n}}{\sqrt{\mu_0 \epsilon_0}} = c\hat{n}$$

$$\langle S \rangle = \langle U \rangle c \hat{n}$$

Energy flux = energy density $\times$ velocity of light

Plane Electromagnetic waves in an isotropic Dielectric medium or in non-conducting medium

$\rho = 0$, $J = \sigma E = 0$

$D = \epsilon E$, $B = \mu H$

has same properties in all directions is called an isotropic dielectric medium.

$$\Rightarrow \vec{\nabla}\cdot\vec{E} = 0 \quad -(1)$$

$$\vec{\nabla}\cdot\vec{B} = 0 \text{ or } \vec{\nabla}\cdot\vec{H} = 0 \quad -(2)$$

$$\vec{\nabla}\times\vec{E} = -\mu\frac{\partial\vec{H}}{\partial t} \quad -(3)$$

$$\vec{\nabla}\times\vec{H} = \epsilon\frac{\partial\vec{E}}{\partial t} \quad -(4)$$

By taking curl of (3) & (4) equations and by using $\nabla\cdot E = 0$, $\nabla\cdot H = 0$

$$\nabla^2 E - \mu\epsilon\frac{\partial^2 E}{\partial t^2} = 0$$

$$\Rightarrow \nabla^2 E - \mu\epsilon\frac{\partial^2 E}{\partial t^2} = 0$$

$$\Rightarrow \nabla^2 H - \mu\epsilon\frac{\partial^2 H}{\partial t^2} = 0$$

$$\Rightarrow \nabla^2\phi - \mu\epsilon\frac{\partial^2\phi}{\partial t^2} = 0 \quad -(5)$$

or general wave equation

$$\nabla^2\phi - \frac{1}{v^2}\frac{\partial^2\phi}{\partial t^2} = 0 \quad -(6)$$

from (5) & (6)

$$\Rightarrow v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_r\mu_0\epsilon_0\epsilon_r}}$$

$\mu_r = \frac{\mu}{\mu_0}$ or $\epsilon_r = \frac{\epsilon}{\epsilon_0}$

$$v = \frac{c}{\sqrt{\mu_r\epsilon_r}}$$

$\mu_r > 1$

$\epsilon_r > 1$

$$\boxed{v < c}$$

As we know refractive index

$$n = \frac{c}{v} \quad \text{i.e.} \quad v = \frac{c}{n}$$

$$n = \sqrt{\mu_r\epsilon_r}$$

for non magnetic material

$$n = \sqrt{\epsilon_r} \Rightarrow \boxed{n^2 = \epsilon_r}$$

for $\nabla^2 E - \frac{1}{v^2}\frac{\partial^2 E}{\partial t^2} = 0$

$\nabla^2 H - \frac{1}{v^2}\frac{\partial^2 H}{\partial t^2} = 0$

Solution $E(r,t) = E_0\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$

$H(r,t) = H_0\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$

$\vec{K} = k\cdot\hat{n} = \frac{2\pi}{\lambda}\cdot\hat{n} = \frac{2\pi\nu}{v}\cdot\hat{n} = \frac{\omega}{v}\cdot\hat{n}$

Or implies $\vec{\nabla}\cdot\vec{E} = 0$ or $\vec{\nabla}\cdot\vec{H} = 0$

$i\,K\cdot E = 0$ $\quad i\,K\cdot H = 0$

$K \perp E \,\&\, H$

Similarly

$\vec{\nabla}\times\vec{E} = i\,\vec{K}\times\vec{E} = -\mu\frac{\partial H}{\partial t} = -\mu(-i\omega H)$

$$\boxed{\vec{K}\times\vec{E} = \mu\omega H}$$

or $\boxed{\vec{K}\times\vec{H} = -\epsilon\omega E}$

wave impedance Z

$Z = \left|\frac{E}{H}\right| = \sqrt{\frac{\mu}{\epsilon}}$

$H = \frac{1}{\mu\omega}\vec{K}\times E = \frac{k\,\hat{n}\times\vec{E}}{\mu\omega} = \frac{\omega}{\mu\,\omega\, v} = \frac{1}{\mu v}(\hat{n}\times E)$

$= \frac{\sqrt{\mu\epsilon}}{\mu}(\hat{n}\times\vec{E})$

$= \sqrt{\frac{\epsilon}{\mu}}\,(\hat{n}\times\vec{E})$

Poynting vector.

$\vec{S} = \vec{E}\times\vec{H} = \frac{E^2}{Z}\hat{n}$